MULTI-WINNER VOTING:

REPRESENTATION, MANIPULATION, DELIBERATION

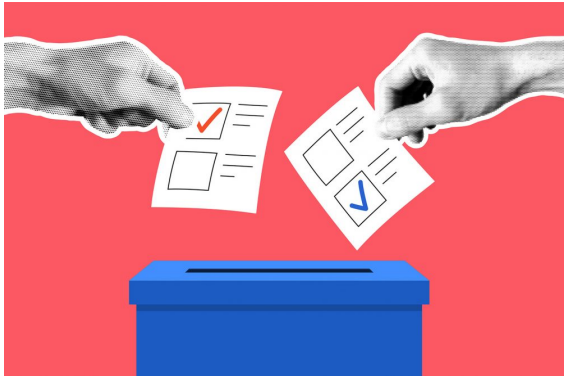
PhD Defense of Oliviero **Nardi**

DBAI, **TU Wien** – 16 October, 2026

Available at tinyurl.com/oliviero-defense

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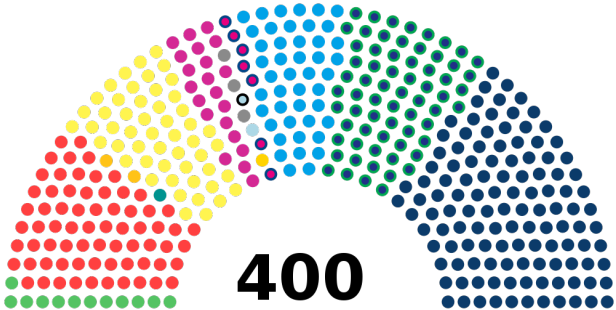


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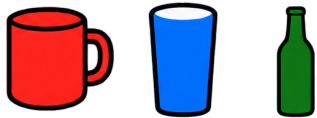
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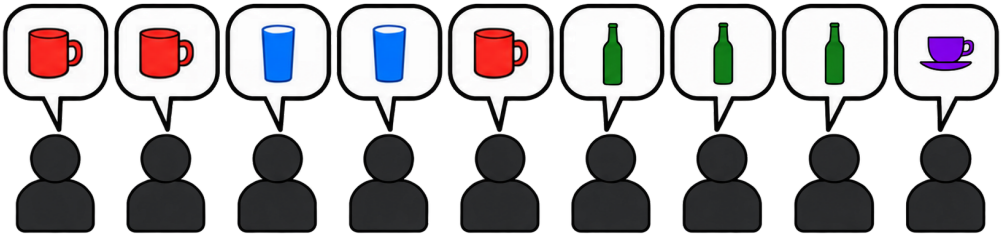
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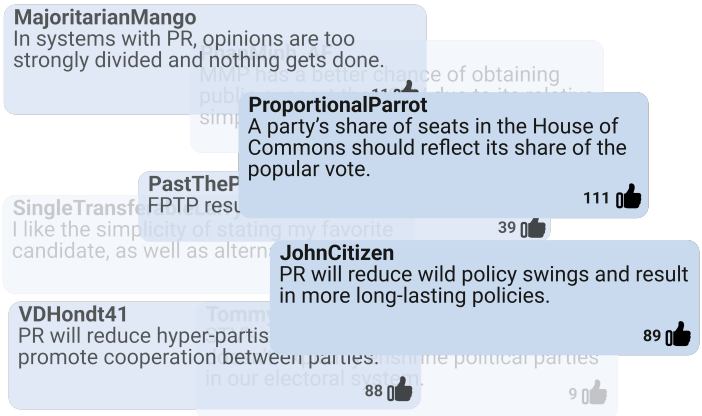


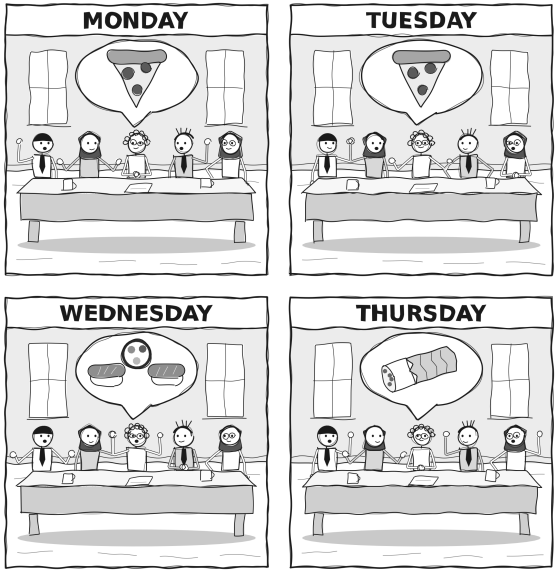

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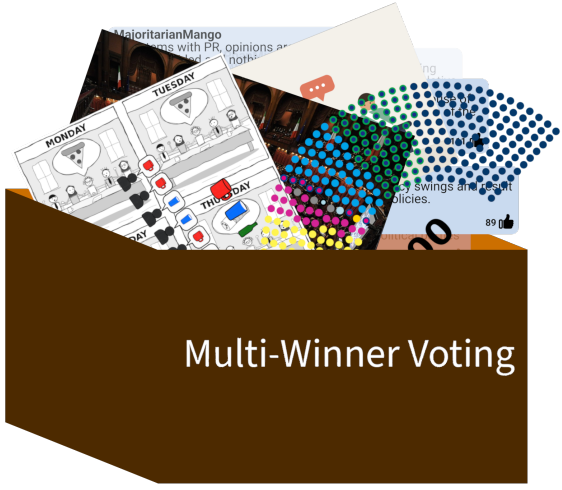







MajoritarianMango
In systems with PR, opinions are too strongly divided and nothing gets done.
ProportionalParrot
A party's share of seats in the House of Commons should reflect its share of the popular vote.
111
39
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PR will reduce wild policy swings and result in more long-lasting policies.
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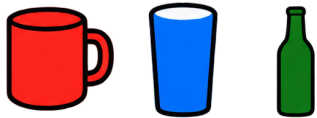

MONDAY
TUESDAY
WEDNESDAY
THURSDAY

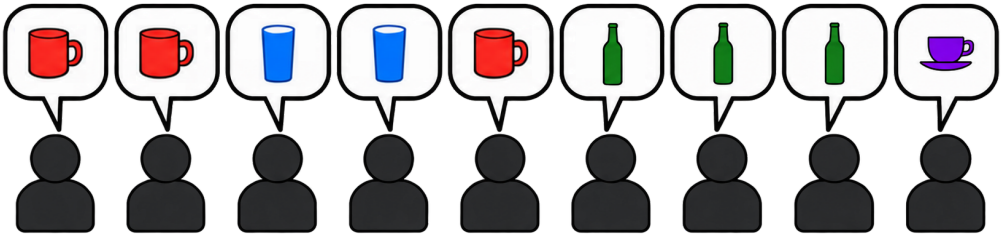

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Multi-Winner Voting

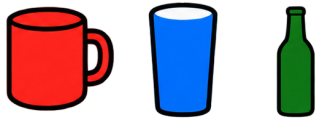
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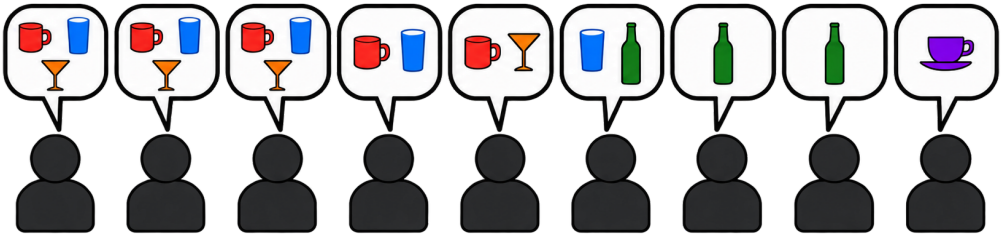
Representation, Manipulation, Deliberation

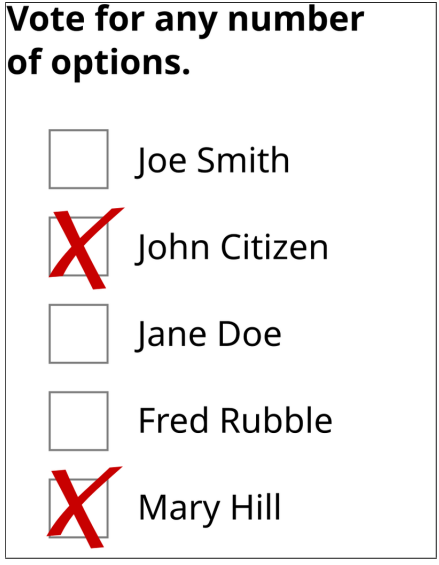
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Representation, Manipulation, Deliberation

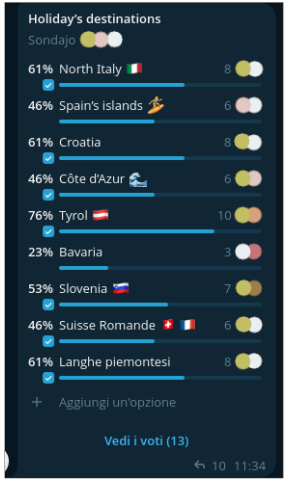






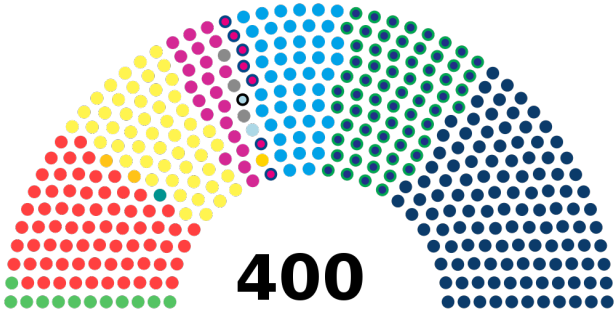



Vote for any number of options.
Joe Smith
John Citizen
Jane Doe
Fred Rubble
Mary Hill


Holiday's destinations
Sondajo
61% North Italy 8
46% Spain's islands 6
61% Croatia 8
46% Côte d'Azur 6
76% Tyrol 10
23% Bavaria 3
53% Slovenia 7
46% Suisse Romande 6
61% Langhe piemontesi 8
Aggiungi un'opzione
Vedi i voti (13)
10 11:34

(Approval-Based)
Multi-Winner Voting:
Representation, Manipulation, Deliberation

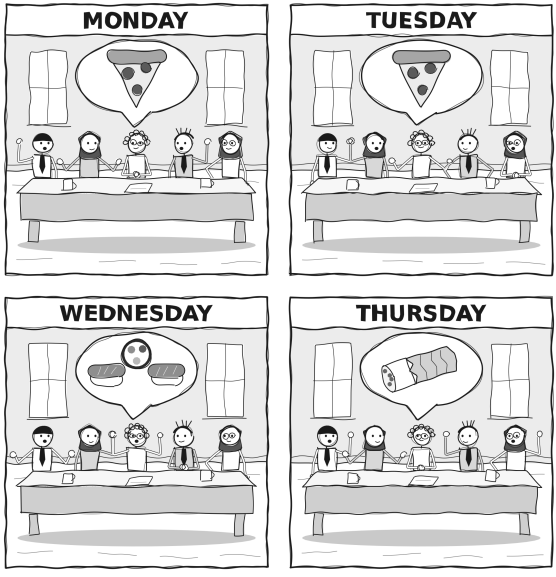
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Multi-Winner Voting:
Representation, Manipulation, Deliberation


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(Approval-Based)

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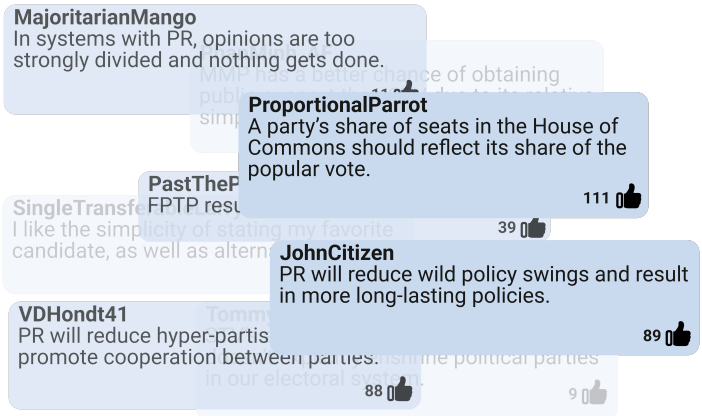
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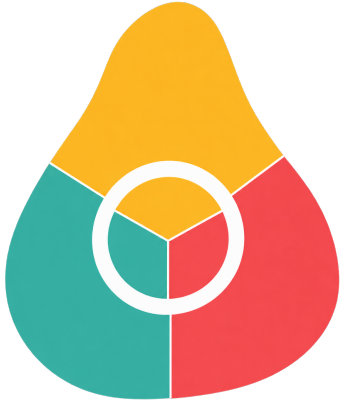
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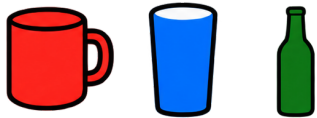

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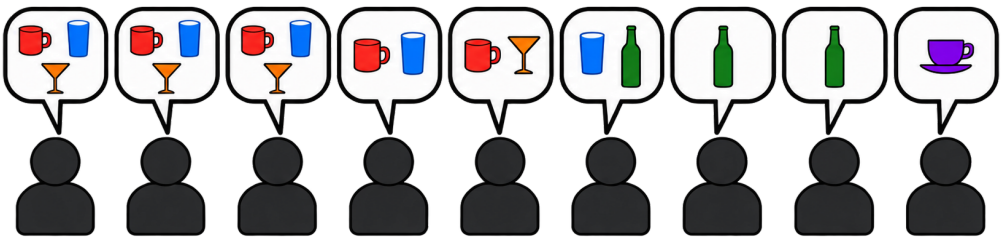
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Multi-Winner Voting:
Representation, Manipulation, Deliberation

1. REPRESENTATION



COMMITTEE VOTING, INFORMALLY





COMMITTEE VOTING, FORMALLY

- Voters $N = \{1, 2, \ldots, n\}$
- Alternatives $C = \{a, b, c, \ldots\}$
- Target committee size $k \in \mathbb{N}$
- Each voter $i \in N$ submits an approval ballot $A_i \subseteq C$
- Collectively, we have a profile of preferences $\boldsymbol{A} = (A_1, A_2, \ldots, A_n)$
- A voting rule F maps each profile $\boldsymbol{A}$ to a winning committee $W \subseteq C$ with $|W| = k$

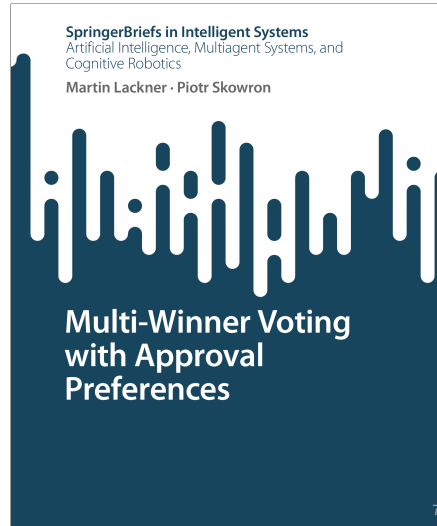
Additionally, we define $N(c) = \{i \in N : c \in A_i\}$.

THE STANDARD CONCEPTION: AVOIDING UNDERREPRESENTATION

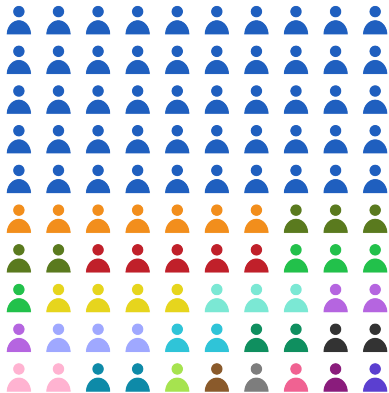
‘if a cohesive group of voters makes up *x*% of the population,
then it should get at least *x*% of the seats’

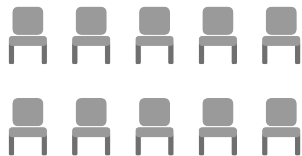
AVOIDING UNDERREPRESENTATION: STATE OF THE ART

- Well-understood and formalised
  - Extended Justified Representation (EJR)
  - EJR+
  - Priceability
- Natural and appealing rules provably satisfy it
  - Proportional Approval Voting
  - Phragmén's Rule
  - Method of Equal Shares
- We argue: Not the last word on representation!

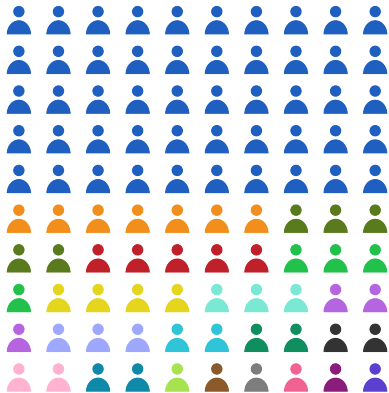


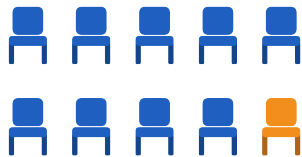
A PROBLEM WITH ‘REPRESENTATIVE’ RULES





A PROBLEM WITH ‘REPRESENTATIVE’ RULES





ANOTHER CONCEPTION: AVOIDING OVERREPRESENTATION

‘if a cohesive group of voters makes up *x*% of the population,
then it should not decide more than *x*% of the seats’

WHY AVOIDING OVERREPRESENTATION?

- Fairness:
  - Prevents groups from having ‘too much’ power
  - Avoids ‘envy’
- Might prevent strategic ‘bullet voting’
- Technical applications
  - Safety in blockchain technologies

N. Boehmer, M. Brill, A. Cevallos, J. Gehrlein, L. Sánchez-Fernández, and U. Schmidt-Kraepelin. Approval-Based Committee Voting in Practice: A Case Study of (Over-)Representation in the Polkadot Blockchain. In *Proceedings of the 38th AAAI Conference on Artificial Intelligence (AAAI-2024)*, pages 9519–9527. AAAI Press, 2024.

FORMALISING OVERREPRESENTATION

‘if a cohesive group of voters makes up x% of the population,
then it should not decide more than x% of the seats’

OVERREPRESENTATION: TENTATIVE DEFINITION

In a winning committee W, for every $c \in W$,
the group of voters $N(c)$ should not have
$|A_i \cap W| > \left\lceil k \frac{|N(c)|}{n} \right\rceil$ for all $i \in N(c)$.

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THE DEFINITION HAS A BUG

Say we want to select $k = 6$ candidates. Consider the following profile:

- $A_1 = \{a, b, c\}$
- $A_2 = \{d, e, f\}$
- $A_3 = \{g\}$

Each voter can get at most $\left\lceil 6 \cdot \frac{1}{3} \right\rceil = 2$ candidates. Any committee overrepresents some voter:

- $\{a, b, c, d, e, f\}$ overrepresents voter 1 and voter 2
- $\{a, b, c, d, e, g\}$ overrepresents voter 1
- etc...

JUSTIFIED UPPER QUOTA (JUQ)

In a winning committee W, for every $c \in W$, if the group of voters $N(c)$ has

$$|A_i \cap W| > \left\lceil k \frac{|N(c)|}{n} \right\rceil \quad \text{for all } i \in N(c),$$

then there cannot be $d \in C \setminus W$ and $N' \subseteq N(d)$ such that

$$|A_i \cap W \setminus \{c\} \cup \{d\}| \leq \left\lceil k \frac{|N'|}{n} \right\rceil \quad \text{for all } i \in N'.$$

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$$|A_i \cap W \setminus \{c\} \cup \{d\}| \leq \left\lceil k\frac{|N'|}{n} \right\rceil \quad \text{for all } i \in N'.$$

JUQ FIXES THE BUG...

Let $k = 6$ and consider again:

- $A_1 = \{a, b, c\}$
- $A_2 = \{d, e, f\}$
- $A_3 = \{g\}$

Recall: each voter can get at most $\lceil 6 \cdot \frac{1}{3} \rceil = 2$ candidates.

Committee $\{a, b, c, d, e, g\}$ **satisfies** JUQ:

- Voter 1 is overrepresented, must kick out c
- But: voter 2 already at quota
- Thus, no need to swap c with f

...BUT NOT TRIVIALLY SO!

Let $k = 6$ and consider again:

- $A_1 = \{a, b, c\}$
- $A_2 = \{d, e, f\}$
- $A_3 = \{g\}$

Recall: each voter can get at most $\left\lceil 6 \cdot \frac{1}{3} \right\rceil = 2$ candidates.

Committee $\{a, b, c, d, e, f\}$ **fails** JUQ:

- Voter 1 is overrepresented, must kick out c
- Voter 3 is underrepresented
- Thus, must swap c with g

WHICH RULES SATISFY JUQ?

Definition

Given a non-increasing $\omega\colon \mathbb{N} \to \mathbb{R}_{\geq 0}$, the ω-Thiele rule picks a committee maximising

$$\sum_{i \in N} \sum_{\ell=1}^{|A_i \cap W|} \omega(\ell)$$

Examples:

- Approval Voting: $\omega(\ell) = (1, 1, 1, \ldots)$
- Proportional Approval Voting: $\omega(\ell) = \left(1, \frac{1}{2}, \frac{1}{3}, \ldots\right)$
- Chamberlin-Courant: $\omega(\ell) = (1, 0, 0, \ldots)$

TERRIFYING NEWS

Proposition

No Thiele rule satisfies JUQ.

EXTENDING THIELE RULES

We define composite Thiele rules, which work in stages of tiebreaking:

Definition

Given a sequence of non-increasing $\omega_1, \omega_2, \ldots : \mathbb{N} \to \mathbb{R}_{\geq 0}$, the corresponding composite Thiele rule picks a committee as follows:

1. First, select all committees maximising the score of ω_1;
2. Among those, select all committees maximising the score of ω_2;
3. Repeat until you are left with a single committee (or without scoring functions).

Theorem

There is a unique composite Thiele rule that satisfies JUQ: we called it Adams Approval Voting.

ADAMS APPROVAL VOTING: DEFINITION

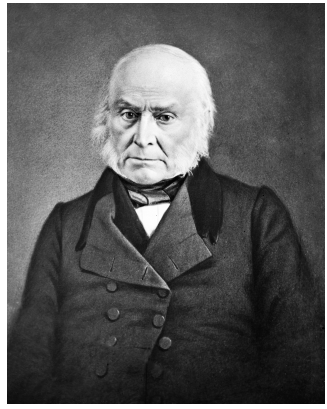
The rule is named after the sixth president of the United States, John Quincy Adams, and the electoral formula he proposed.

Definition

Adams Approval Voting is the composite Thiele rule defined by

1. $\omega_1(\ell) = (1, 0, 0, 0, 0, \ldots)$ and
2. $\omega_2(\ell) = \left(1, 1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \ldots\right)$.

We adapt it from parliamentary elections to committee voting.



ADAMS APPROVAL VOTING: EXPLANATION

Concretely, the rule proceeds as follows:

1. Select all committees W maximising the number of 'covered' voters, i.e., that maximise

$$\left|\{i \in N : A_i \cap W \neq \varnothing\}\right|$$

2. Among those, select all committees W' that maximise

$$\sum_{i \in N} \sum_{\ell=2}^{|A_i \cap W'|} \frac{1}{\ell - 1}$$

OTHER RESULTS: OVERREPRESENTATION

- We show that one can find a JUQ committee in polynomial time
- Another perspective on representation: balancing over- and underrepresentation
  - We introduce another axiom, justified near quota
  - Analogous results: Only Sainte-Laguë-AV satisfies it
- Relationship between our axioms and established underrepresentation axioms (e.g., EJR+)
  - We show that some natural avenues in proving compatibility fail

... And many other results!

2. MANIPULATION



SEQUENTIAL VOTING

- Voters $N = \{1, 2, \ldots, n\}$ (fixed throughout)
- Pre-set number of rounds $k \in \mathbb{N}$:
  - Alternatives C_1
  - Alternatives C_2
  - ...
  - Alternatives C_k
- For each round r, every voter $i \in N$ submits an approval ballot $A_i^r \subseteq C_r$

SEQUENTIAL VOTING

- Voters $N = \{1, 2, \ldots, n\}$ (fixed throughout)
- Pre-set number of rounds $k \in \mathbb{N}$:
  - Alternatives $C_1 = \{a, b, c, \ldots\}$
  - Alternatives $C_2 = \{▲, ■, ⬟, \ldots\}$
  - ...
  - Alternatives $C_k = \{●, ●, ●, \ldots\}$
- For each round r, every voter $i \in N$ submits an approval ballot $A_i^r \subseteq C_r$

SEQUENTIAL VOTING: MAJORITY RULE

Voters	**Decision 1**	**Decision 2**	**Decision 3**	**Decision 4**
Alice				
Rabbit				
Hatter				
Carroll				
Winner				

SEQUENTIAL VOTING: MAJORITY RULE

Voters	**Decision 1**	**Decision 2**	**Decision 3**	**Decision 4**
Alice	$\{a, b\}$			
Rabbit	$\{a, c\}$			
Hatter	$\{a\}$			
Carroll	$\{b\}$			
Winner				

SEQUENTIAL VOTING: MAJORITY RULE

Voters	**Decision 1**	**Decision 2**	**Decision 3**	**Decision 4**
Alice	$\{a, b\}$			
Rabbit	$\{a, c\}$			
Hatter	$\{a\}$			
Carroll	$\{b\}$			
Winner	a			

SEQUENTIAL VOTING: MAJORITY RULE

Voters	**Decision 1**	**Decision 2**	**Decision 3**	**Decision 4**
Alice	$\{a, b\}$	$\{b\}$		
Rabbit	$\{a, c\}$	$\{b, c\}$		
Hatter	$\{a\}$	$\{a, b\}$		
Carroll	$\{b\}$	$\{a, c\}$		
Winner	a			

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Voters	**Decision 1**	**Decision 2**	**Decision 3**	**Decision 4**
Alice	$\{a, b\}$	$\{b\}$		
Rabbit	$\{a, c\}$	$\{b, c\}$		
Hatter	$\{a\}$	$\{a, b\}$		
Carroll	$\{b\}$	$\{a, c\}$		
Winner	a	b		

SEQUENTIAL VOTING: MAJORITY RULE

Voters	**Decision 1**	**Decision 2**	**Decision 3**	**Decision 4**
Alice	$\{a, b\}$	$\{b\}$	$\{a, c\}$	
Rabbit	$\{a, c\}$	$\{b, c\}$	$\{b, c\}$	
Hatter	$\{a\}$	$\{a, b\}$	$\{c\}$	
Carroll	$\{b\}$	$\{a, c\}$	$\{a, b\}$	
Winner	a	b		

SEQUENTIAL VOTING: MAJORITY RULE

Voters	**Decision 1**	**Decision 2**	**Decision 3**	**Decision 4**
Alice	$\{a, b\}$	$\{b\}$	$\{a, c\}$	
Rabbit	$\{a, c\}$	$\{b, c\}$	$\{b, c\}$	
Hatter	$\{a\}$	$\{a, b\}$	$\{c\}$	
Carroll	$\{b\}$	$\{a, c\}$	$\{a, b\}$	
Winner	a	b	c	

SEQUENTIAL VOTING: MAJORITY RULE

Voters	**Decision 1**	**Decision 2**	**Decision 3**	**Decision 4**
Alice	$\{a, b\}$	$\{b\}$	$\{a, c\}$	$\{a, b\}$
Rabbit	$\{a, c\}$	$\{b, c\}$	$\{b, c\}$	$\{a, c\}$
Hatter	$\{a\}$	$\{a, b\}$	$\{c\}$	$\{a\}$
Carroll	$\{b\}$	$\{a, c\}$	$\{a, b\}$	$\{b, c\}$
Winner	a	b	c	

SEQUENTIAL VOTING: MAJORITY RULE

Voters	**Decision 1**	**Decision 2**	**Decision 3**	**Decision 4**
Alice	$\{a, b\}$	$\{b\}$	$\{a, c\}$	$\{a, b\}$
Rabbit	$\{a, c\}$	$\{b, c\}$	$\{b, c\}$	$\{a, c\}$
Hatter	$\{a\}$	$\{a, b\}$	$\{c\}$	$\{a\}$
Carroll	$\{b\}$	$\{a, c\}$	$\{a, b\}$	$\{b, c\}$
Winner	a	b	c	a

SEQUENTIAL VOTING: ‘REPRESENTATIVE’ RULE

Voters	Weight	Decision 1	Decision 2	Decision 3	Decision 4
Alice	1				
Rabbit	1				
Hatter	1				
Carroll	1				
Winner					

SEQUENTIAL VOTING: 'REPRESENTATIVE' RULE

Voters	**Weight**	**Decision 1**	**Decision 2**	**Decision 3**	**Decision 4**
Alice	1	$\{a, b\}$			
Rabbit	1	$\{a, c\}$			
Hatter	1	$\{a\}$			
Carroll	1	$\{b\}$			
Winner					

SEQUENTIAL VOTING: 'REPRESENTATIVE' RULE

Voters	**Weight**	**Decision 1**	**Decision 2**	**Decision 3**	**Decision 4**
Alice	0.5	$\{a, b\}$			
Rabbit	0.5	$\{a, c\}$			
Hatter	0.5	$\{a\}$			
Carroll	1	$\{b\}$			
Winner		a			

SEQUENTIAL VOTING: 'REPRESENTATIVE' RULE

Voters	**Weight**	**Decision 1**	**Decision 2**	**Decision 3**	**Decision 4**
Alice	0.5	$\{a, b\}$	$\{b\}$		
Rabbit	0.5	$\{a, c\}$	$\{b, c\}$		
Hatter	0.5	$\{a\}$	$\{a, b\}$		
Carroll	1	$\{b\}$	$\{a, c\}$		
Winner		a			

SEQUENTIAL VOTING: ‘REPRESENTATIVE’ RULE

Voters	**Weight**	**Decision 1**	**Decision 2**	**Decision 3**	**Decision 4**
Alice	0.33	$\{a, b\}$	$\{b\}$		
Rabbit	0.33	$\{a, c\}$	$\{b, c\}$		
Hatter	0.33	$\{a\}$	$\{a, b\}$		
Carroll	1	$\{b\}$	$\{a, c\}$		
Winner		a	b		

SEQUENTIAL VOTING: ‘REPRESENTATIVE’ RULE

Voters	**Weight**	**Decision 1**	**Decision 2**	**Decision 3**	**Decision 4**
Alice	0.33	$\{a, b\}$	$\{b\}$	$\{a, c\}$	
Rabbit	0.33	$\{a, c\}$	$\{b, c\}$	$\{b, c\}$	
Hatter	0.33	$\{a\}$	$\{a, b\}$	$\{c\}$	
Carroll	1	$\{b\}$	$\{a, c\}$	$\{a, b\}$	
Winner		a	b		

SEQUENTIAL VOTING: ‘REPRESENTATIVE’ RULE

Voters	**Weight**	**Decision 1**	**Decision 2**	**Decision 3**	**Decision 4**
Alice	0.25	$\{a, b\}$	$\{b\}$	$\{a, c\}$	
Rabbit	0.33	$\{a, c\}$	$\{b, c\}$	$\{b, c\}$	
Hatter	0.33	$\{a\}$	$\{a, b\}$	$\{c\}$	
Carroll	0.5	$\{b\}$	$\{a, c\}$	$\{a, b\}$	
Winner		a	b	a	

SEQUENTIAL VOTING: ‘REPRESENTATIVE’ RULE

Voters	**Weight**	**Decision 1**	**Decision 2**	**Decision 3**	**Decision 4**
Alice	0.25	$\{a, b\}$	$\{b\}$	$\{a, c\}$	$\{a, b\}$
Rabbit	0.33	$\{a, c\}$	$\{b, c\}$	$\{b, c\}$	$\{a, c\}$
Hatter	0.33	$\{a\}$	$\{a, b\}$	$\{c\}$	$\{a\}$
Carroll	0.5	$\{b\}$	$\{a, c\}$	$\{a, b\}$	$\{b, c\}$
Winner		a	b	a	

SEQUENTIAL VOTING: ‘REPRESENTATIVE’ RULE

Voters	**Weight**	**Decision 1**	**Decision 2**	**Decision 3**	**Decision 4**
Alice	0.2	$\{a, b\}$	$\{b\}$	$\{a, c\}$	$\{a, b\}$
Rabbit	0.25	$\{a, c\}$	$\{b, c\}$	$\{b, c\}$	$\{a, c\}$
Hatter	0.25	$\{a\}$	$\{a, b\}$	$\{c\}$	$\{a\}$
Carroll	0.5	$\{b\}$	$\{a, c\}$	$\{a, b\}$	$\{b, c\}$
Winner		a	b	a	a

OUR MAIN NEGATIVE POINT

Minimal forms of representative guarantees
already allow for very simple forms of manipulation:
FREE-RIDING!

FREE-RIDING: EXAMPLE

Voters	Weight	Decision 1	Decision 2	Decision 3
Alice	1			
Rabbit	1			
Hatter	1			
Carroll	1			
Winner				

FREE-RIDING: EXAMPLE

Voters	**Weight**	**Decision 1**	**Decision 2**	**Decision 3**
Alice	1	$\{a\}$		
Rabbit	1	$\{a\}$		
Hatter	1	$\{a\}$		
Carroll	1	$\{a\}$		
Winner				

FREE-RIDING: EXAMPLE

Voters	**Weight**	**Decision 1**	**Decision 2**	**Decision 3**
Alice	0.5	$\{a\}$		
Rabbit	0.5	$\{a\}$		
Hatter	0.5	$\{a\}$		
Carroll	0.5	$\{a\}$		
Winner		a		

FREE-RIDING: EXAMPLE

Voters	**Weight**	**Decision 1**	**Decision 2**	**Decision 3**
Alice	0.5	$\{a\}$	$\{a\}$	
Rabbit	0.5	$\{a\}$	$\{a\}$	
Hatter	0.5	$\{a\}$	$\{a\}$	
Carroll	0.5	$\{a\}$	$\{b\}$	
Winner		a		

FREE-RIDING: EXAMPLE

Voters	**Weight**	**Decision 1**	**Decision 2**	**Decision 3**
Alice	0.33	$\{a\}$	$\{a\}$	
Rabbit	0.33	$\{a\}$	$\{a\}$	
Hatter	0.33	$\{a\}$	$\{a\}$	
Carroll	0.5	$\{a\}$	$\{b\}$	
Winner		a	a	

FREE-RIDING: EXAMPLE

Voters	**Weight**	**Decision 1**	**Decision 2**	**Decision 3**
Alice	0.33	$\{a\}$	$\{a\}$	$\{a\}$
Rabbit	0.33	$\{a\}$	$\{a\}$	$\{a\}$
Hatter	0.33	$\{a\}$	$\{a\}$	$\{b\}$
Carroll	0.5	$\{a\}$	$\{b\}$	$\{c\}$
Winner		a	a	

FREE-RIDING: EXAMPLE

Voters	**Weight**	**Decision 1**	**Decision 2**	**Decision 3**
Alice	0.25	$\{a\}$	$\{a\}$	$\{a\}$
Rabbit	0.25	$\{a\}$	$\{a\}$	$\{a\}$
Hatter	0.33	$\{a\}$	$\{a\}$	$\{b\}$
Carroll	0.5	$\{a\}$	$\{b\}$	$\{c\}$
Winner		a	a	a

FREE-RIDING: EXAMPLE

Voters	Weight	Decision 1	Decision 2	Decision 3
Alice	1			
Rabbit	1			
Hatter	1			
Carroll	1			
Winner				

FREE-RIDING: EXAMPLE

Voters	**Weight**	**Decision 1**	**Decision 2**	**Decision 3**
Alice	1	$\{a\}$		
Rabbit	1	$\{a\}$		
Hatter	1	$\{a\}$		
Carroll	1	$\{b\}$		
Winner				

FREE-RIDING: EXAMPLE

Voters	**Weight**	**Decision 1**	**Decision 2**	**Decision 3**
Alice	0.5	$\{a\}$		
Rabbit	0.5	$\{a\}$		
Hatter	0.5	$\{a\}$		
Carroll	1	$\{b\}$		
Winner		a		

FREE-RIDING: EXAMPLE

Voters	**Weight**	**Decision 1**	**Decision 2**	**Decision 3**
Alice	0.5	$\{a\}$	$\{a\}$	
Rabbit	0.5	$\{a\}$	$\{a\}$	
Hatter	0.5	$\{a\}$	$\{a\}$	
Carroll	1	$\{b\}$	$\{b\}$	
Winner		a		

FREE-RIDING: EXAMPLE

Voters	**Weight**	**Decision 1**	**Decision 2**	**Decision 3**
Alice	0.33	$\{a\}$	$\{a\}$	
Rabbit	0.33	$\{a\}$	$\{a\}$	
Hatter	0.33	$\{a\}$	$\{a\}$	
Carroll	1	$\{b\}$	$\{b\}$	
Winner		a	a	

FREE-RIDING: EXAMPLE

Voters	**Weight**	**Decision 1**	**Decision 2**	**Decision 3**
Alice	0.33	$\{a\}$	$\{a\}$	$\{a\}$
Rabbit	0.33	$\{a\}$	$\{a\}$	$\{a\}$
Hatter	0.33	$\{a\}$	$\{a\}$	$\{b\}$
Carroll	1	$\{b\}$	$\{b\}$	$\{c\}$
Winner		a	a	

FREE-RIDING: EXAMPLE

Voters	**Weight**	**Decision 1**	**Decision 2**	**Decision 3**
Alice	0.25	$\{a\}$	$\{a\}$	$\{a\}$
Rabbit	0.25	$\{a\}$	$\{a\}$	$\{a\}$
Hatter	0.33	$\{a\}$	$\{a\}$	$\{b\}$
Carroll	0.5	$\{b\}$	$\{b\}$	$\{c\}$
Winner		a	a	c

OUR MAIN POSITIVE POINT

FREE-RIDING can backfire!

FREE-RIDING CAN BACKFIRE

Voters	Weight	Decision 1	Decision 2	Decision 3	Decision 4
Alice	1				
Rabbit	1				
Hatter	1				
Carroll	1				
Winner					

FREE-RIDING CAN BACKFIRE

Voters	Weight	Decision 1	Decision 2	Decision 3	Decision 4
Alice	1	$\{b\}$			
Rabbit	1	$\{a\}$			
Hatter	1	$\{a\}$			
Carroll	1	$\{a\}$			
Winner					

FREE-RIDING CAN BACKFIRE

Voters	**Weight**	**Decision 1**	**Decision 2**	**Decision 3**	**Decision 4**
Alice	1	$\{b\}$			
Rabbit	0.5	$\{a\}$			
Hatter	0.5	$\{a\}$			
Carroll	0.5	$\{a\}$			
Winner		a			

FREE-RIDING CAN BACKFIRE

Voters	**Weight**	**Decision 1**	**Decision 2**	**Decision 3**	**Decision 4**
Alice	1	$\{b\}$	$\{a\}$		
Rabbit	0.5	$\{a\}$	$\{b\}$		
Hatter	0.5	$\{a\}$	$\{c\}$		
Carroll	0.5	$\{a\}$	$\{b\}$		
Winner		a			

FREE-RIDING CAN BACKFIRE

Voters	**Weight**	**Decision 1**	**Decision 2**	**Decision 3**	**Decision 4**
Alice	0.5	$\{b\}$	$\{a\}$		
Rabbit	0.5	$\{a\}$	$\{b\}$		
Hatter	0.5	$\{a\}$	$\{c\}$		
Carroll	0.5	$\{a\}$	$\{b\}$		
Winner		a	a		

FREE-RIDING CAN BACKFIRE

Voters	**Weight**	**Decision 1**	**Decision 2**	**Decision 3**	**Decision 4**
Alice	0.5	$\{b\}$	$\{a\}$	$\{b\}$	
Rabbit	0.5	$\{a\}$	$\{b\}$	$\{b\}$	
Hatter	0.5	$\{a\}$	$\{c\}$	$\{a\}$	
Carroll	0.5	$\{a\}$	$\{b\}$	$\{a\}$	
Winner		a	a		

FREE-RIDING CAN BACKFIRE

Voters	**Weight**	**Decision 1**	**Decision 2**	**Decision 3**	**Decision 4**
Alice	0.5	$\{b\}$	$\{a\}$	$\{b\}$	
Rabbit	0.5	$\{a\}$	$\{b\}$	$\{b\}$	
Hatter	0.33	$\{a\}$	$\{c\}$	$\{a\}$	
Carroll	0.33	$\{a\}$	$\{b\}$	$\{a\}$	
Winner		a	a	a	

FREE-RIDING CAN BACKFIRE

Voters	**Weight**	**Decision 1**	**Decision 2**	**Decision 3**	**Decision 4**
Alice	0.5	$\{b\}$	$\{a\}$	$\{b\}$	$\{b\}$
Rabbit	0.5	$\{a\}$	$\{b\}$	$\{b\}$	$\{a\}$
Hatter	0.33	$\{a\}$	$\{c\}$	$\{a\}$	$\{b\}$
Carroll	0.33	$\{a\}$	$\{b\}$	$\{a\}$	$\{a\}$
Winner		a	a	a	

FREE-RIDING CAN BACKFIRE

Voters	**Weight**	**Decision 1**	**Decision 2**	**Decision 3**	**Decision 4**
Alice	0.5	$\{b\}$	$\{a\}$	$\{b\}$	$\{b\}$
Rabbit	0.33	$\{a\}$	$\{b\}$	$\{b\}$	$\{a\}$
Hatter	0.33	$\{a\}$	$\{c\}$	$\{a\}$	$\{b\}$
Carroll	0.25	$\{a\}$	$\{b\}$	$\{a\}$	$\{a\}$
Winner		a	a	a	a

FREE-RIDING CAN BACKFIRE

Voters	**Weight**	**Decision 1**	**Decision 2**	**Decision 3**	**Decision 4**
Alice	0.5	$\{b\}$	$\{a\}$	$\{b\}$	$\{b\}$
Rabbit	0.33	$\{a\}$	$\{b\}$	$\{b\}$	$\{a\}$
Hatter	0.33	$\{a\}$	$\{c\}$	$\{a\}$	$\{b\}$
Carroll	0.25	$\{a\}$	$\{b\}$	$\{a\}$	$\{a\}$
Winner		a	a	a	a

FREE-RIDING CAN BACKFIRE

Voters	Weight	Decision 1	Decision 2	Decision 3	Decision 4
Alice	1				
Rabbit	1				
Hatter	1				
Carroll	1				
Winner					

FREE-RIDING CAN BACKFIRE

Voters	Weight	Decision 1	Decision 2	Decision 3	Decision 4
Alice	1	$\{b\}$			
Rabbit	1	$\{a\}$			
Hatter	1	$\{a\}$			
Carroll	1	$\{c\}$			
Winner					

FREE-RIDING CAN BACKFIRE

Voters	**Weight**	**Decision 1**	**Decision 2**	**Decision 3**	**Decision 4**
Alice	1	$\{b\}$			
Rabbit	0.5	$\{a\}$			
Hatter	0.5	$\{a\}$			
Carroll	1	$\{c\}$			
Winner		a			

FREE-RIDING CAN BACKFIRE

Voters	**Weight**	**Decision 1**	**Decision 2**	**Decision 3**	**Decision 4**
Alice	1	$\{b\}$	$\{a\}$		
Rabbit	0.5	$\{a\}$	$\{b\}$		
Hatter	0.5	$\{a\}$	$\{c\}$		
Carroll	1	$\{c\}$	$\{b\}$		
Winner		a			

FREE-RIDING CAN BACKFIRE

Voters	**Weight**	**Decision 1**	**Decision 2**	**Decision 3**	**Decision 4**
Alice	1	$\{b\}$	$\{a\}$		
Rabbit	0.33	$\{a\}$	$\{b\}$		
Hatter	0.5	$\{a\}$	$\{c\}$		
Carroll	0.5	$\{c\}$	$\{b\}$		
Winner		a	b		

FREE-RIDING CAN BACKFIRE

Voters	**Weight**	**Decision 1**	**Decision 2**	**Decision 3**	**Decision 4**
Alice	1	$\{b\}$	$\{a\}$	$\{b\}$	
Rabbit	0.33	$\{a\}$	$\{b\}$	$\{b\}$	
Hatter	0.5	$\{a\}$	$\{c\}$	$\{a\}$	
Carroll	0.5	$\{c\}$	$\{b\}$	$\{a\}$	
Winner		a	b		

FREE-RIDING CAN BACKFIRE

Voters	**Weight**	**Decision 1**	**Decision 2**	**Decision 3**	**Decision 4**
Alice	0.5	$\{b\}$	$\{a\}$	$\{b\}$	
Rabbit	0.25	$\{a\}$	$\{b\}$	$\{b\}$	
Hatter	0.5	$\{a\}$	$\{c\}$	$\{a\}$	
Carroll	0.5	$\{c\}$	$\{b\}$	$\{a\}$	
Winner		a	b	b	

FREE-RIDING CAN BACKFIRE

Voters	**Weight**	**Decision 1**	**Decision 2**	**Decision 3**	**Decision 4**
Alice	0.5	$\{b\}$	$\{a\}$	$\{b\}$	$\{b\}$
Rabbit	0.25	$\{a\}$	$\{b\}$	$\{b\}$	$\{a\}$
Hatter	0.5	$\{a\}$	$\{c\}$	$\{a\}$	$\{b\}$
Carroll	0.5	$\{c\}$	$\{b\}$	$\{a\}$	$\{a\}$
Winner		a	b	b	

FREE-RIDING CAN BACKFIRE

Voters	**Weight**	**Decision 1**	**Decision 2**	**Decision 3**	**Decision 4**
Alice	0.33	$\{b\}$	$\{a\}$	$\{b\}$	$\{b\}$
Rabbit	0.25	$\{a\}$	$\{b\}$	$\{b\}$	$\{a\}$
Hatter	0.33	$\{a\}$	$\{c\}$	$\{a\}$	$\{b\}$
Carroll	0.5	$\{c\}$	$\{b\}$	$\{a\}$	$\{a\}$
Winner		a	b	b	b

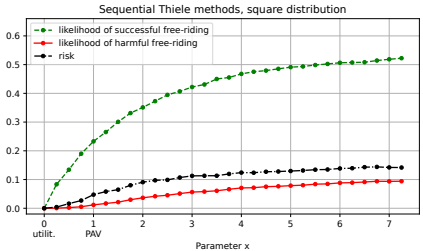
FREE-RIDING CAN BACKFIRE

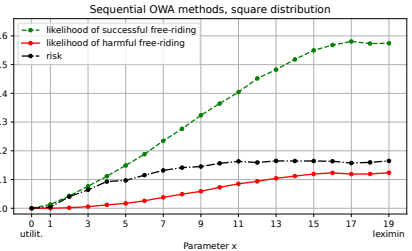
Voters	**Weight**	**Decision 1**	**Decision 2**	**Decision 3**	**Decision 4**
Alice	0.33	$\{b\}$	$\{a\}$	$\{b\}$	$\{b\}$
Rabbit	0.25	$\{a\}$	$\{b\}$	$\{b\}$	$\{a\}$
Hatter	0.33	$\{a\}$	$\{c\}$	$\{a\}$	$\{b\}$
Carroll	0.5	$\{c\}$	$\{b\}$	$\{a\}$	$\{a\}$
Winner		a	b	b	b

SUMMARY: MANIPULATION

Using this fact, we show:

- For a large class of rules, it is computationally hard to decide if free-riding is beneficial
- That the risk of free-riding is non-neglibile (via simulations)

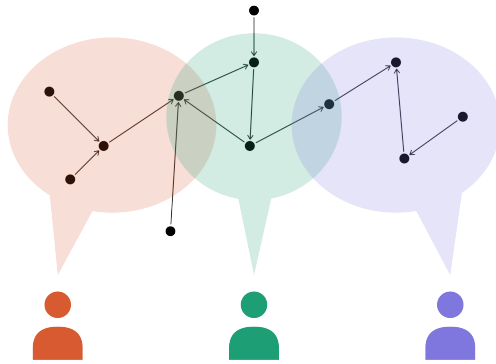




MESSAGE

FREE-RIDING is unavoidable in ‘fair’ sequential voting,
but it is risky and its consequences hard to predict! ✓

3. DELIBERATION



SUMMARISING DISCUSSIONS

PastThePost

FPTP results in more stable governance.

39

MajoritarianMango

In systems with PR, opinions are too strongly divided and nothing gets done.

11

VDHondt41

PR will reduce hyper-partisanship and promote cooperation between parties.

88

ProportionalParrot

A party's share of seats in the House of Commons should reflect its share of the popular vote.

111

JohnCitizen

PR will reduce wild policy swings and result in more long-lasting policies.

89

SingleTransferableLarry

I like the simplicity of stating my favorite candidate, as well as alternative choices.

8

PhanMinh_AF

MMP has a better chance of obtaining public support than STV due to its relative simplicity.

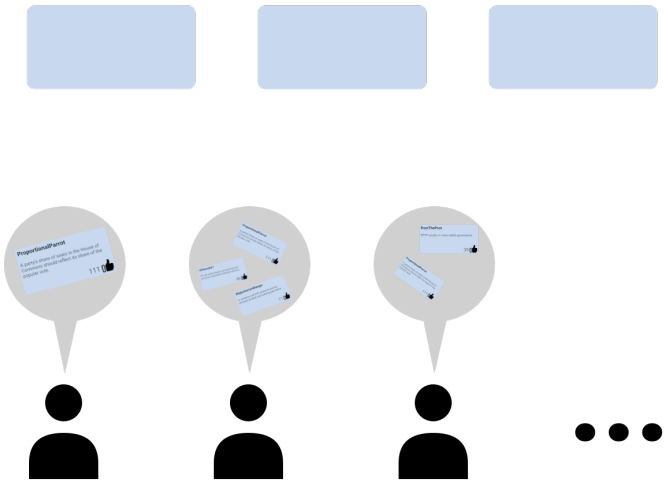
6

TommyHill

STV's advantage over MMP is that it doesn't explicitly enshrine political parties in our electoral system.

9

SUMMARISING VIA APPROVAL VOTING



SUMMARISING VIA APPROVAL VOTING

ProportionalParrot

A party's share of seats in the House of Commons should reflect its share of the popular vote.

111

PastThePost

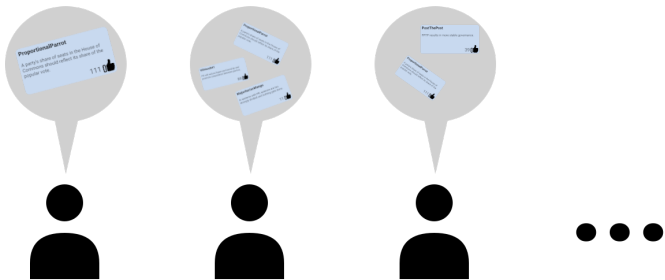
FPTP results in more stable governance.

39

PhanMinh_AF

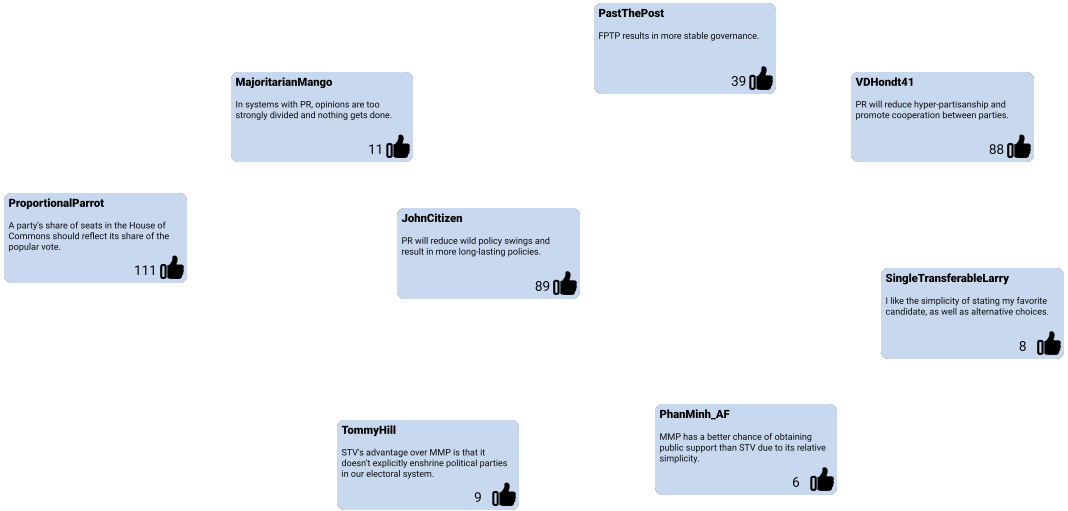
MMP has a better chance of obtaining public support than STV due to its relative simplicity.

6

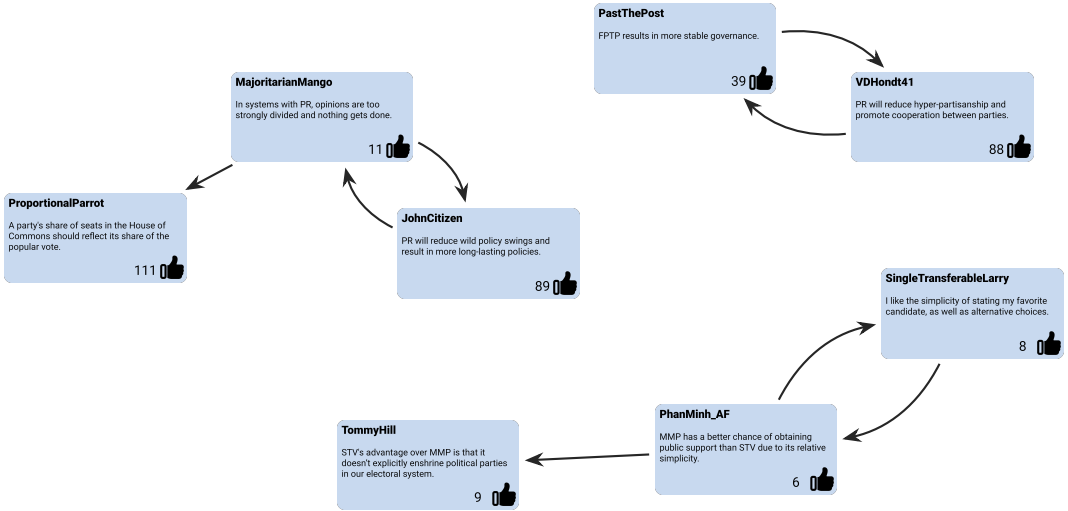


Ignores the broader argumentation!

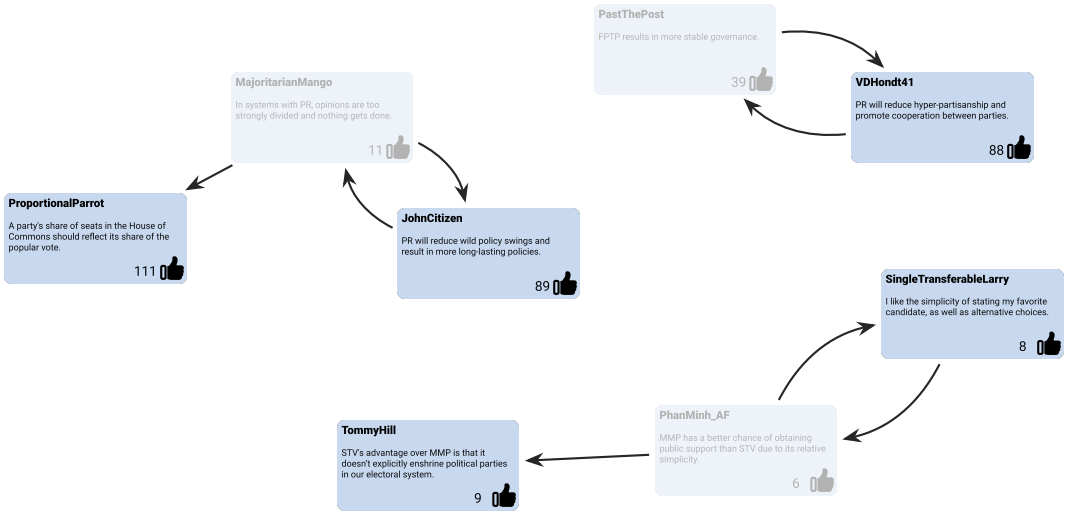
SUMMARISING VIA ARGUMENTATIVE STRUCTURE



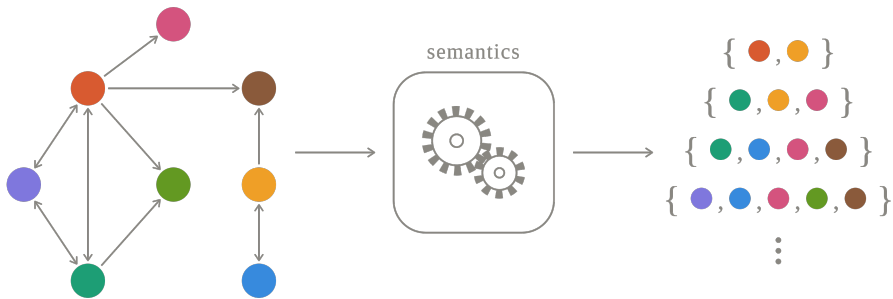
SUMMARISING VIA ARGUMENTATIVE STRUCTURE



SUMMARISING VIA ARGUMENTATIVE STRUCTURE

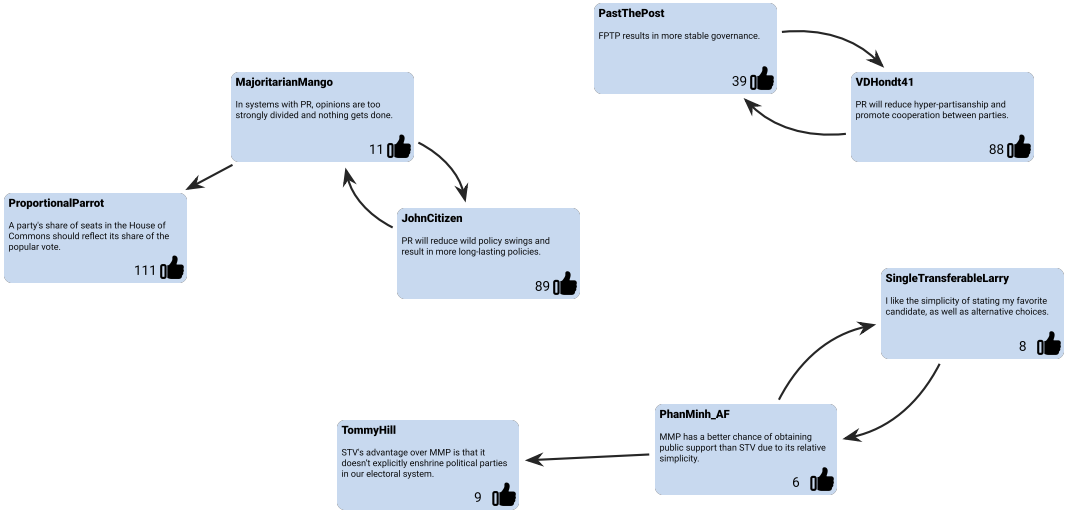


ABSTRACT ARGUMENTATION

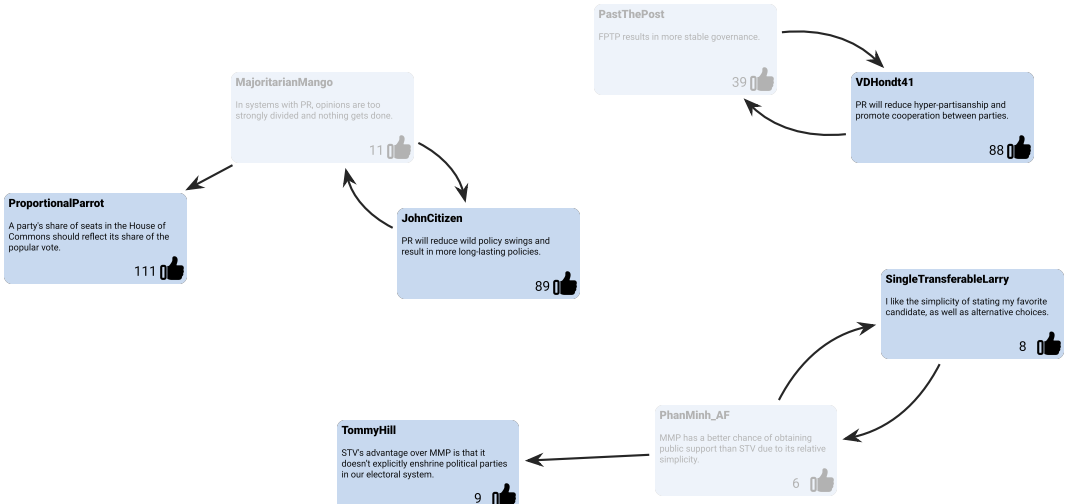


P. M. Dung. On the Acceptability of Arguments and its Fundamental Role in Nonmonotonic Reasoning, Logic Programming and *n*-Person Games. *Artificial Intelligence*, 77(2):321–358, 1995.

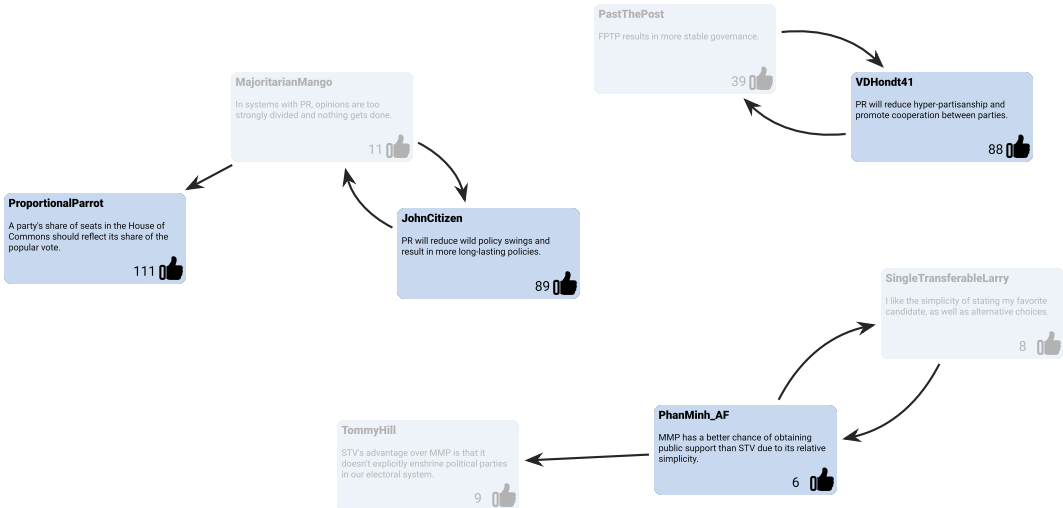
COUNTING THE EXTENSIONS



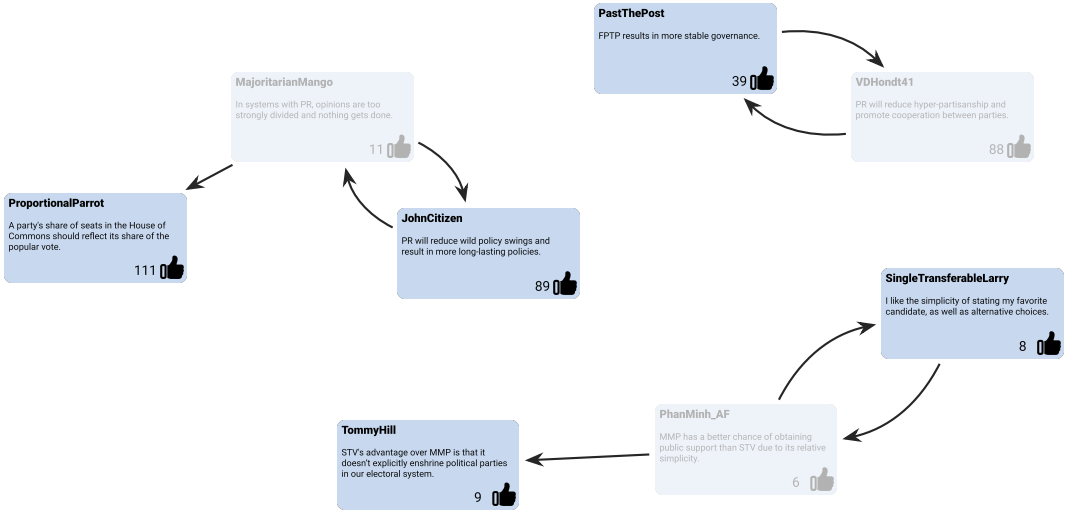
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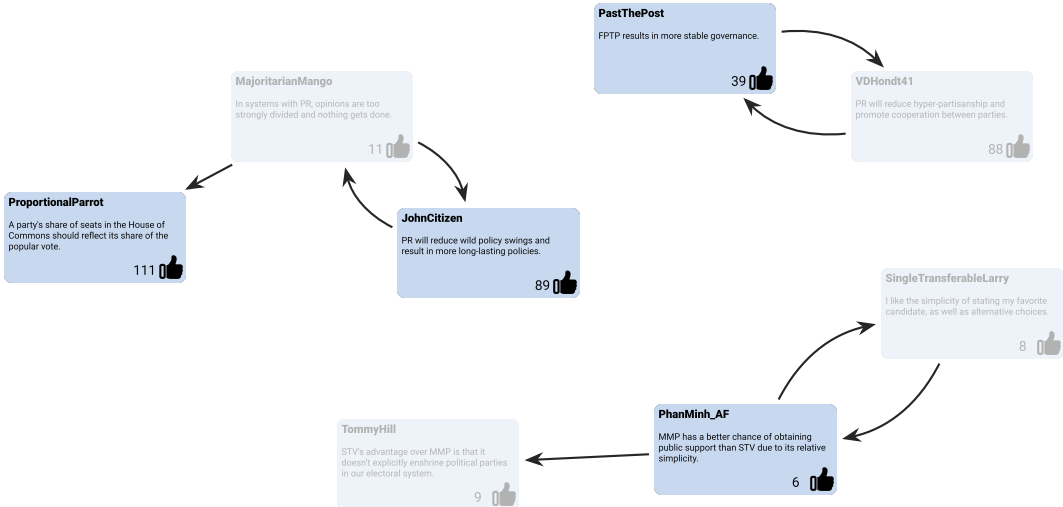
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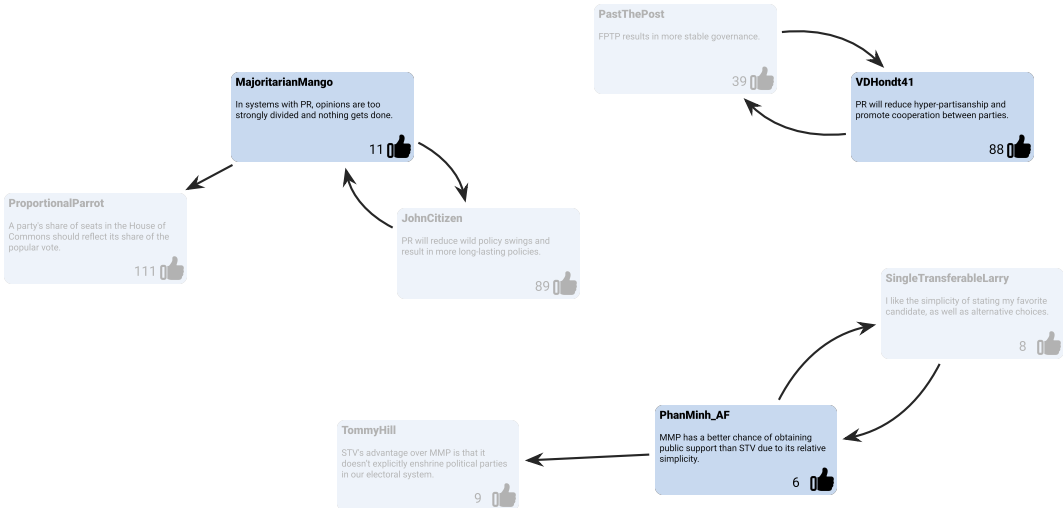
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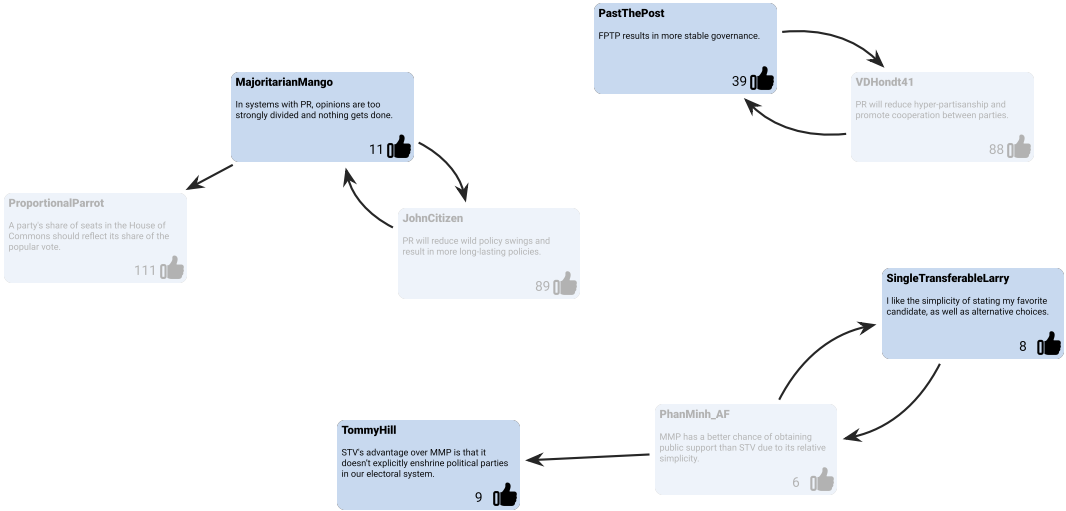
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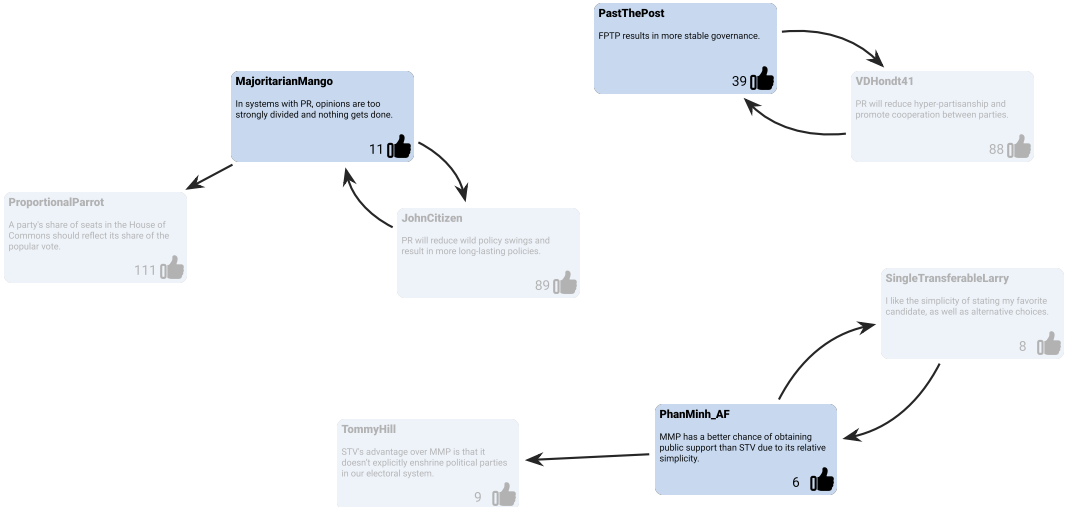
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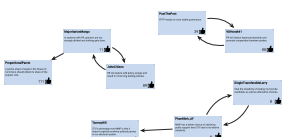


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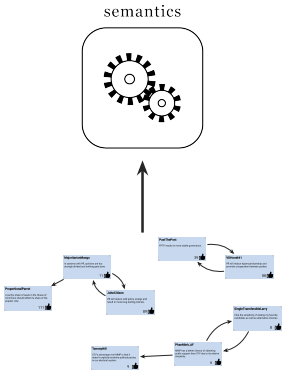


Abstract argumentation can be verbose!

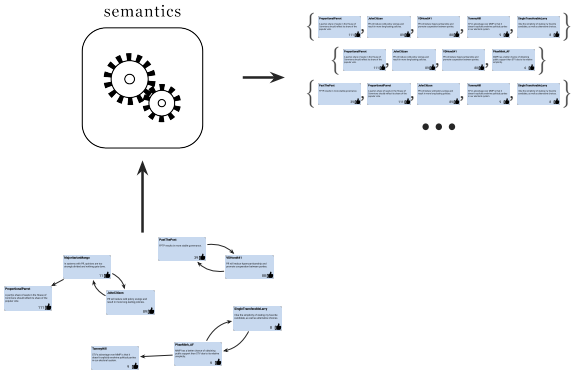
COMBINING ARGUMENTATION AND VOTING



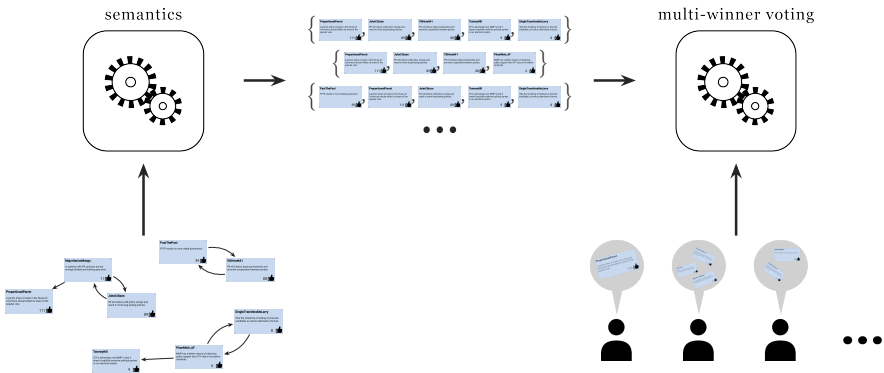
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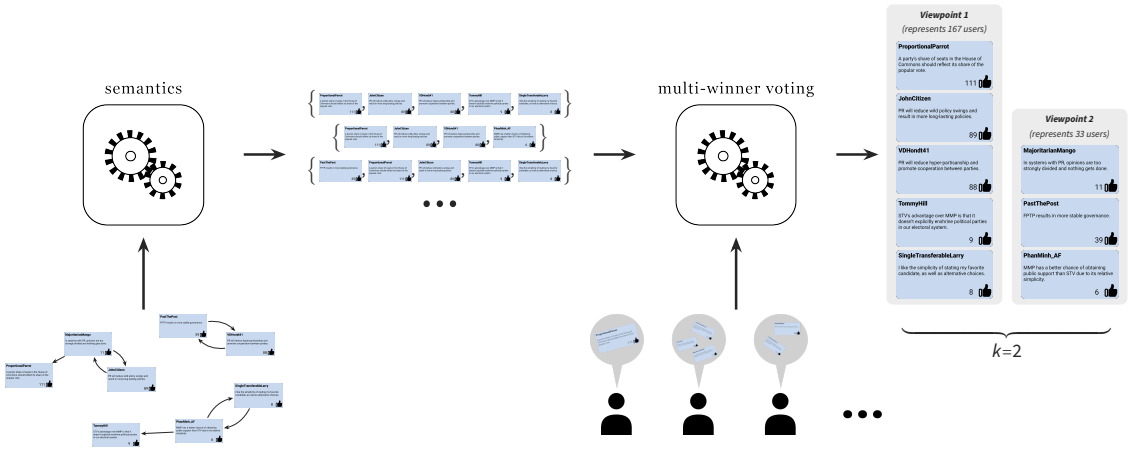
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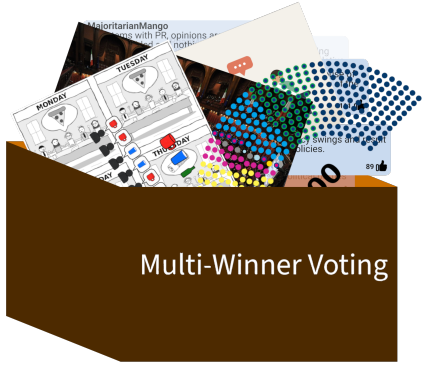
COMBINING ARGUMENTATION AND VOTING



SUMMARY: DELIBERATION

- We define a formal model of discussions (argumentation framework + approvals)
  - Outcomes are sets of extensions filtered using the preferences
- We quantify the representativeness of an outcome
- We introduce outcome-selection algorithms based on multi-winner voting ideas
- We study the rules computationally, axiomatically and experimentally

CONCLUSION



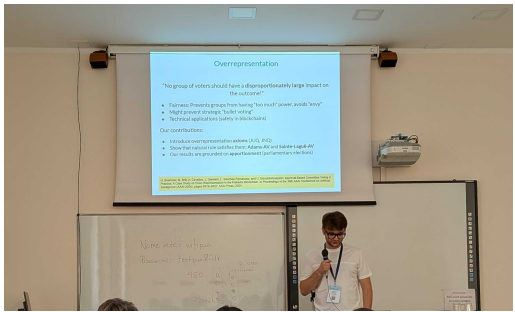
RECAP

- Representation:
  - We introduced formal definitions of overrepresentation in committee voting, along with rules to prevent it.
- Manipulation:
  - We studied free-riding in sequential voting, and show that it is not as appealing a strategy as one might think.
- Deliberation:
  - We introduce an approach for summarising online discussions, combining argumentation and voting theory.

FINAL SLIDE



First conference in Vienna



Last conference in Rome